Sentinel - The Autonomous System Health Guardian (v2.0)

1. The Scenario

Sentinel is an intelligent monitoring agent system designed for a cloud-native platform. Beyond investigating system anomalies in real-time, *Sentinel* must also act as the platform's primary **Data Intelligence Interface**, allowing system administrators to query massive operational datasets using natural language.

2. Core Requirements

A. Multi-Agent System (CrewAI & Specialized Roles)

Students must define **four** specialized agents:

- **Monitor Agent:** Parses logs and telemetry to detect anomalies.
- **Diagnostician Agent:** Uses a Knowledge Base (RAG) to determine root causes.
- **Remediation Agent:** Proposes/executes fixes and performs verification.
- **Data Intelligence Agent:** Interfaces with the SQL database to answer natural language queries regarding infrastructure health.

B. The Database Infrastructure & NL2SQL

- **The Data Pack:** Students must connect to the provided system_telemetry.db (containing servers, logs, and incidents tables).
- **NL2SQL Interface:** Create an agent capable of translating natural language queries into SQL.
- **Graceful Degradation:** If a user queries information outside the scope of the schema (e.g., "What is the CEO's favorite color?"), the agent must identify this limitation and respond gracefully rather than hallucinating or crashing.

C. The Workflow (LangGraph Circular Loop)

Students must use **LangGraph** to create a self-healing circular workflow:

- **Monitor $\rightarrow$ Diagnose $\rightarrow$ Remediate $\rightarrow$ Verify.**
- If *Verify* fails, the loop triggers a "Retry" flag and returns to *Diagnose*.

D. MCP Implementation (The Standardized Gateway)

The MCP Server must act as the **only** entry point for agents to interact with the system. It must expose:

- get_system_logs()
- execute_system_command() (Read-only/Simulated)
- query_knowledge_base()
- query_database(nl_query) (Must interact with the SQL DB)

E. Production Resilience

- **Safety Guardrails:** Remediation Agent must prevent "high-risk" commands (e.g., system_reboot) during critical database load.
- **Observability:** All steps must be logged into **LangSmith**. Students must submit a screenshot of their trace for an NL2SQL query to demonstrate how the model translated intent to code.

3. Deliverables

1. **The "Sentinel" Repository:** Fully modularized codebase, including the MCP Server, the LangGraph orchestrator, and the SQL database connector.
2. **The "Post-Mortem" Report:** A technical report simulating an infrastructure outage. Document how the agents identified the issue, the SQL queries used to investigate, and the final mitigation results.
3. **Visual Workflow:** A diagram representing the LangGraph state transitions and the agent hierarchy.